

ICT102: Recursive Algorithms Assignment

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course :ict 102

Data structures and algorithms

PART A: Algorithm Design

1. Factorial Algorithm

The factorial algorithm calculates the product of all positive integers up to n using recursion. It breaks the problem down by multiplying n by the result of $(n-1)!$

Logic: $n! = n \times (n - 1)!$

Base Case: if $n = 0$, return 1

2. Fibonacci Algorithm

This algorithm finds the n th number in the Fibonacci sequence by summing the two preceding numbers. It utilizes multiple recursion.

Logic: $F(n) = F(n - 1) + F(n - 2)$

Base Cases: if $n = 0$, return 0; if $n = 1$, return 1

PART B: Pseudocode

Factorial Pseudocode

```
FUNCTION factorial (n)
    IF n is 0 THEN
        RETURN 1
    ELSE
        RETURN n* factorial(n - 1)
END FUNCTION
```

Fibonacci Pseudocode

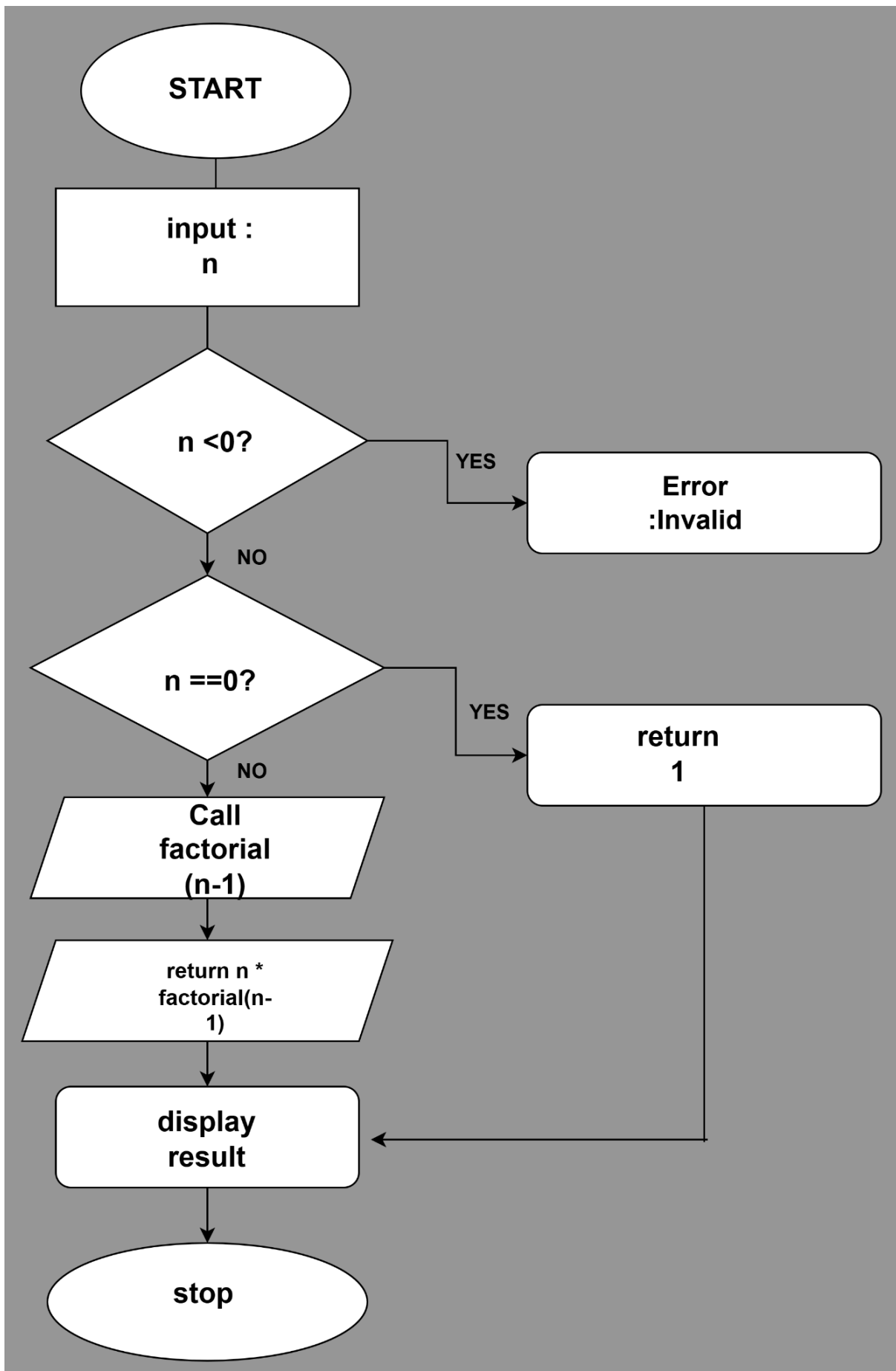
```
FUNCTION fibonacci(n)
    IF n is 0 THEN RETURN 0
    IF n is 1 THEN RETURN 1
    ELSE
        RETURN fibonacci( n - 2)
END FUNCTION
```

PART C:Python Source Code

```
def factorial(n):
    if n < 0:
        raise ValueError("Factorial is not defined for negative numbers.")
```

```
    if n == 0:
        return 1
    return n * factorial(n - 1)
def fibonacci(n):
    if n < 0:
        raise ValueError("Fibonacci is not defined for negative
numbers.")
    if n == 0:
        return 0
    if n == 1:
        return 1
    return fibonacci(n - 1) + fibonacci(n - 2)
# Main Execution Logic
if __name__ == "__main__":
    n = 5 # Example test
    print(f"Factorial of {n}: {factorial(n)}")
    print(f"Fibonacci of {n}: {fibonacci(n)}")
```

FLOW CHART FOR FACTORIAL SEQUENCE



FIBONACCI FLOW CHART

